TECHNICAL SPECIFICATION OF EQUIPMENT FOR ONLINE EFFLUENT PARAMETER MEASUREMENT (pH, BOD, COD & TSS)

GENERAL REQUIREMENT:

- Analyser should be able to produce analytical valid results and flag sensor fault.
- Sensors should be robust and rugged and is capable of operating in variable and extreme measurement condition, and still maintaining its calibrated status.
- The entire monitoring station including the sensors and hardware should be tamper proof.
- The analyser should record all operation information in log files.
- Should have non volatile data logger for recording at least one year continuous data.
- Should have facility of plant level data viewing and retrieval with selection of Ethernet, wireless, Modbus and USB.
- Records of calibration and validation should be automatically uploaded on web server from NAPS to UPPCB / CPCB.
- Online diagnostic features should be available.
- System should have expandable program to calculate parameter load daily, weekly or monthly basis for future evaluation with flow rate signal output.
- BOD/COD measurement should be based on the Total organic Carbon (TOC) measurement / UV Visible spectroscopy double beam complying to US EPA 415.1 / 415.2 or equivalent standard for effluent/sewage/waste water.
- The seamless data to be transferred to CPCB and UPPCB server through 2G – GSM on continuous basis for seamless data transmission.
- The Server and software should be as per CPCB protocol C 12011/33/2015 Tech dated 11.06.2015. (Copy enclosed)
- The equipment should be TUV, MCERT, USEPA certified
- Certificate from OEM for authorization of data management.
- Every system supplier will comply with testing/calibration protocol as per international standards of the CPCB.

The major prerequisites of efficient online analysers are:

- Should be capable of operating unattended over prolonged period of time.
- Should produce analytically valid results with precision and repeatability to transmit validated data to CPCB/UPPCB server.
- The instrument/analyser should be robust and rugged, for optimal operation under extreme environmental conditions, while maintaining its calibrated status.
- The analyser should have inbuilt features for automatic water matrix change adaption.
- The instrument / analyser should have onboard library of calibration spectras for different industrial matrices with provision of accumulating further calibration matrices.
- Should have provision for Multi-server data transmission from each station without intermediate PC or plant server.

- Should have provision to send system alarm to central server in case any changes made in configuration or calibration.
- Should have provision to record all operation information in log file.
- For each parameter there should be provision for independent analysis, validation, calibration & data transmission.
- For TOC analyser, the empirical relationship between TOC to COD or BOD must be authenticated for all industrial applications and the correlation calculation (for factor) provided.
- Record of calibration and validation should be available on real time basis on central server from each location/parameter.
- Record of online diagnostic features including sensor status should be available in database for user friendly maintenance.
- Must have low operation and maintenance requirements with low chemical consumption and recurring cost of consumables and spares.

Quantification:

- Instrument Calibration: The following frequency has to be used for calibration of analyser
- pH – once every week or as specified by manufacturer whichever is earlier
- COD – once every week or as specified by manufacturer whichever is earlier
- BOD – once every week or as specified by manufacturer whichever is earlier
- TSS – once every week or as specified by manufacturer whichever is earlier

The software must keep all calibration data points in memory for interpretation of matrix change adaption.

System Validation:

Online instrument operation will be evaluated using the known buffers, traceable standards and laboratory techniques. By validating sensors and probes with known standards such as KHP (potassium hydrogen phthalate) for COD & TOC, Formazin equivalent standard for TSS & pH buffers have to be used to calculate a running variance of the measurements. When the variance is outside of the set points, this can be an indication the monitor requires calibration and service.

Parameter validation:

Each parameter is validated with reference to standard laboratory analysis and known standards.

Parameter Accuracy: Allowed Variability:

The relative difference between online and laboratory measurements has to be between

- COD Accuracy ±10% or better.
- BOD Accuracy ±10% or better.
- pH Accuracy ±0.2 pH or better.
- TSS Accuracy +/-10 % or better.

Operation & maintenance:

- Daily Check – GPS Transmission, System Diagnostic alarms.
- Monthly Check – Sensors & system cleaning, data backup, Parameter Calibration as specified in calibration schedule.
- Periodic Check – System validation with known standards, Laboratory & Online parameters Comparative.

Reporting:

The suppliers have to provide central server at CPCB and SPCB with latest software to view the data in graphical/ tabular format and also to compare the data features.
One minute data average must be transmitted/retrieved to servers every 30 minutes. In the event of transmission loss the time stamped data in the datalogger memory must be transmitted to fill from the last transmission break with a stamp of time delay. The software should have two way communication, so that data from the system can be seen whenever desired and remote of controller/data logger can be taken to visualize the immediate status of the system.

Data Management:

Considering the heterogeneity of real time monitoring systems, NAPS is required to submit real time data through their respective instrument suppliers. This mechanism will help in consolidating the data avoiding the complexity of different technologies and availability of monitored data in different data formats and at the same time involving the instrument suppliers in data transferring mechanism. The system enables two way communication required to manage such real time systems.

The basic functional capabilities of such software systems shall include:-

- The system should be capable of collecting data on real time basis without any human intervention.
- The data generation, data pick up, data transmission; data integration at server end should be automatic.
- The submitted data shall be available to the Boards, SPCBs and CPCB for immediate corrective action.
- Raw data should be transmitted simultaneously to SPCBs and CPCB.
- Configurations of the systems once set up (through remote procedure) and verified, should not be changed. In case any setting change is required it should be notified and recorded through the authorized representatives only.

- The data submitted electronically shall be available to the data generator through internet, so that corrective action if any required due to submission of erroneous data can be initiated by the industry.
- The software should be capable to verify the data correctness which means at any given point of time the regulatory authorities/data generator should be able to visualize the current data of any location's specific parameter.
- A system for data validation shall be incorporated in the software with two stage/three stage validation and fixed responsibilities of stakeholders as below;

  a. **Data Generator**:
  b. **SPCBs**:
  c. **CPCB**:

- Change Request Management: window for requesting data changes due to actual field conditions shall be provided at NAPS in line to SPCB to consider the request or not.
- The site surrounding environmental conditions shall also be recorded along with other environmental parameters, as these have the potential to affect the system adversely and corrupt the data generated.
- System should have capability to depict data at the actual location of industry over the map. CPCB and or SPCBs shall develop a map based system for data integration at a single location.

  The software should be capable of analyzing the data with statistical tools and shall have the following capabilities:

i) Statistical data analysis (customizable) for average, min., max., diurnal variation.
ii) Capability of comparison of data with respect to standards/threshold values.
iii) Auto report and , auto mail generation etc.
iv) Providing calibration database for further validation/correction of data.
vi) Providing data in export format on continuous basis through central/station computer system to other system.

- Data Storage for next five years.
- System should be connected to a backup power source with adequate capacity to avoid any power disruption
- Considering the large volume of data required to be collated, compiled, processed and interpreted a software system will be developed in future at CPCB in consultation with all SPCBs/PCCs exploring common data format to collect data from different servers to a common server.

Technical Specification for online COD, BOD, TSS & pH Measurement.

<table>
<tr><td>Analyser Type</td><td colspan="3">Cabinet Type & Multi-parameter</td></tr>
<tr><td>Parameter to be measured</td><td colspan="3">BOD
COD
TSS
pH</td></tr>
<tr><td rowspan="3">Principle of measurement</td><td>COD/BOD</td><td colspan="2">TOC (Total organic carbon measurement) principle / UV-Vis Spectrophotometry (double beam with entire spectrum scanning.</td></tr>
<tr><td>TSS</td><td colspan="2">Light scattering/ reflection principle</td></tr>
<tr><td>pH</td><td colspan="2">Electrochemical principle</td></tr>
<tr><td rowspan="4">Measuring Range</td><td>COD</td><td>0 to 500 mg/lit. Accuracy ± 10%</td><td rowspan="4">Accuracy with reference to laboratory analysis.</td></tr>
<tr><td>BOD</td><td>0 to 200 mg/lit. or higher. Accuracy ± 10%</td></tr>
<tr><td>TSS</td><td>0 to 1000 mg/lit. Accuracy ± 10%</td></tr>
<tr><td>pH</td><td>0 to 14. Accuracy ± 0.2 pH</td></tr>
<tr><td>Operating Pressure</td><td colspan="3">4 Kg/CM^2</td></tr>
<tr><td>Operating Temperature</td><td colspan="3">Sample temperature: 0 ºC to 50 ºC / Ambient temperature – 50^{o}C</td></tr>
<tr><td>Operating Flow</td><td colspan="3">0 - 5 L/MIN</td></tr>
<tr><td>Enclosure protection</td><td colspan="3">IP65 or better</td></tr>
<tr><td>Analog output</td><td colspan="3">4-20 mA output for individual parameter</td></tr>
<tr><td>Communication</td><td colspan="3">-RS232 required (Sub-D 9 ways female connector) with 2 meters cable for PC
-RS485 required for the connection of external probes
-USB port required for USB key connection</td></tr>
<tr><td>Memory</td><td colspan="3">5000 records (up to 16 measurement channels) with date and time</td></tr>
<tr><td>Display</td><td colspan="3">Colour TFT LCD/LED (alpha numeric display of parameter in engineering unit)</td></tr>
<tr><td>Measuring time/Response time</td><td colspan="3">3 minutes or better</td></tr>
<tr><td>Measuring Cycle</td><td colspan="3">Continuous</td></tr>
<tr><td>Power supply</td><td colspan="3">230 VAC 50 Hz</td></tr>
<tr><td>Cleaning method</td><td colspan="3">Automatic</td></tr>
<tr><td>Calibration requirement</td><td colspan="3">Zero Calibration Facility should be there.
Span adjustment either not required or should be adjustable.</td></tr>
<tr><td>Reagent if any</td><td colspan="3">Not required</td></tr>
<tr><td rowspan="9">Additional Features</td><td colspan="3">Analyser should be cabinet type for easy operation, maintenance & troubleshooting.</td></tr>
<tr><td colspan="3">USB port is required for recorded measurement download, screen copy function (easy troubleshooting) & software update</td></tr>
<tr><td colspan="3">Wi-Fi & Ethernet module should be available for remote data transfer.</td></tr>
<tr><td colspan="3">Reagent-free measurement</td></tr>
<tr><td colspan="3">Maintenance Free.</td></tr>
<tr><td colspan="3">Operating cost should be low.</td></tr>
<tr><td colspan="3">Data Transmission from Analyser to PC, CPCB & UPPCB server. Supplier to make available all the measurement data to UPPCB server so that on-line data is available at UPPCB and CPCB server.</td></tr>
<tr><td colspan="3">Auto calibration facility.</td></tr>
<tr><td colspan="3">All the accessories required for mounting the sensor are to be provided.</td></tr>
<tr><td></td><td colspan="3">Self cleaning (automatic)</td></tr>
</table>

Compliance of statuary guidelines issued time to time and issued by CPCB are applicable for compliance up to the date of tendering.